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Where Deer Meet Roads: A five-Year Spatial and Temporal Analysis of Deer-Vehicle Incidents in Montclair, New Jersey

I. Introduction

Deer populations across the eastern United States have expanded significantly over recent decades, bringing deer into increasingly close contact with suburban road networks. In 2004, deer-vehicle collisions (DVCs) were already a significant public safety concern, accounting for more than 1.5 million incidents annually, generating an estimated $1.1 billion in vehicle damage and approximately 150 human fatalities each year (Hedlund et al., 2004). Fast forward to 2022, and the scale of the problem had grown dramatically. Peer-reviewed research published in Current Biology estimated DVCs had risen to approximately 2.1 million annually, generating more than $10 billion in economic losses and resulting in roughly 440 human fatalities and 59,000 injuries each year(Cunningham et al., 2022). By 2024–2025, real-world insurance data confirmed the trend was holding: State Farm and the Insurance Information Institute (III) recorded approximately 1.7 million animal collision insurance claims in that period alone, with deer accounting for more than 1.1 million of those claims, noting that insured claims do not represent all collisions that occur (State Farm, 2025; Insurance Information Institute, 2025).

Research has demonstrated that these incidents are not randomly distributed. Deer-vehicle collisions cluster around specific landscape features, intensify during predictable seasonal windows driven by deer behavioral cycles, and concentrate along roads that intersect wildlife movement corridors (Found & Boyce, 2011; Morelle et al., 2020). Factors such as proximity to forested land, road geometry (curvature, speed limits, sight distance, elevation of a given road), and time of day have all been identified as significant and recurring predictors of collision risk (Hubbard et al., 2004; Morelle et al., 2020). As suburban development continues to expand into historically rural and forested landscapes, the geographic reach of these conflicts has grown alongside it (Hedlund et al., 2004).

The Township of Montclair, New Jersey, reflects this broader pattern in a locally specific and well-documented way. Situated in Essex County along the eastern face of the Watchung Mountain ridgeline, Montclair spans a notable elevation gradient, rising from lower, flatter terrain in its eastern sections near Bloomfield and Glen Ridge up toward the higher western portions bordering West Orange and Verona. This topographic shift reflects the rise of the First Watchung Mountain, a ridgeline running northeast–southwest through Essex County that forms the township's western boundary and defines its relationship to the forested wildlife habitats that lie just beyond it. Directly adjacent to that western edge of Montclair sits Eagle Rock Reservation, a 408-acre forested preserve managed by the Essex County Park System that spans the First Watchung Mountain across West Orange, Montclair, and Verona. To the south, South Mountain Reservation, a 2,112-acre nature reserve extending through West Orange, Maplewood, and Millburn along the first and second Watchung ridges, lies in close proximity to the township as well. Both reservations are documented habitats for white-tailed deer and other wildlife (Essex County Parks, 2025a, 2025b).

It is within this geographic and ecological context that deer-related incidents in Montclair have become a growing and measurable public safety concern. Between 2020 and 2025, the Township recorded 222 confirmed deer carcass removal events, reflecting a 117% increase over the five-year study period. These figures, derived from official municipal invoices issued by the township's Health Department, reflect a pattern with clear spatial and temporal structure. Certain streets account for a disproportionate share of incidents. Certain months produce dramatically higher rates than others. And the overall trajectory, with only one year of decline across the entire study period, points to a problem that is accelerating rather than stabilizing. This study represents the first systematic, GIS-based examination of deer incidents in Montclair, and aims to identify where, when, and why these incidents are occurring, and what the township can do in response.

What makes Montclair a particularly valuable case study is the quality and consistency of its underlying data. Unlike many communities where DVC data is limited to self-reported insurance claims or police reports, sources known to significantly undercount actual incidents. Montclair's deer carcass removal invoices provide a verified, address-level record of every documented deer mortality requiring municipal intervention (Hubbard et al., 2004). Each entry is supported by a physical location, a documented municipal service, and a corresponding financial transaction, creating a layered verification system that substantially reduces reporting bias. This administrative dataset, spanning five consecutive years, offers a rare opportunity to conduct a longitudinal spatial analysis of deer incidents at the municipal scale.

This paper analyzes that data through three primary lenses: temporal trends, examining how incidents have changed year over year and month over month; spatial patterns, identifying where incidents are concentrated and how those concentrations have shifted over time; and policy implications, evaluating what road safety interventions the evidence supports. The findings are intended to serve both as an academic contribution to the growing literature on urban deer management and as a practical resource for the Township of Montclair's Department of Sustainability as it considers data-driven approaches to wildlife conflict mitigation.

II. Data Collection and Preparation

The incident data at the core of this study is derived from a single primary source: The official deer carcass removal invoices processed through the Township of Montclair's Department of Health and Human Services. These invoices document carcass removal services conducted by the township's contracted vendor, Space Farms Zoo & Museum, and serve as administrative records that provide verifiable accounts of every deer-related incident that required municipal intervention. Each invoice contains the exact address of the removal, the date of service, and the associated municipal transaction record, making them a precise and traceable foundation for analysis. For example, the 2021 invoices record individual removals at locations such as Highland Avenue, Upper Mountain Avenue, and Llewellyn Road, with each entry tied to a specific service date (Township of Montclair, 2021). To enable systematic analysis, a compiled dataset was constructed directly from these invoices, resulting in a structured record of 222 confirmed deer incidents spanning 2020 to 2025 (Township of Montclair, 2020–2025).

A major strength of this study lies in the reliability of its underlying data source. Unlike many wildlife studies that rely on self-reported sightings or survey-based observations, this dataset is derived entirely from verified municipal operational records. Each entry corresponds to a confirmed service event, meaning that every incident in the dataset represents an actual deer carcass removal rather than an unverified report. As discussed in the Introduction, this stands in contrast to the self-reported insurance claims and police reports that form the basis of most DVC datasets, sources known to significantly undercount actual incidents (Hubbard et al., 2004). The municipal invoice system adds a further layer of accountability through its three-part verification structure: each incident is tied to a physical location, a documented municipal service, and a corresponding financial transaction. This combination substantially reduces reporting bias and provides a strong empirical foundation for the spatial and temporal analysis that follows.

The transformation of raw invoice records into a usable analytical dataset involved a careful multi-step process. All available invoices were reviewed, including both physical PDF documents and digital purchase order records spanning the full 5-year study period. Relevant information (specifically street addresses, dates of removal, and service details) was manually extracted from each document. Because invoices frequently grouped multiple incidents within a single entry, additional parsing was required to isolate individual records before analysis could begin. Each address was then standardized to ensure consistency in formatting, correcting abbreviations, spelling variations, and incomplete entries where necessary. The cleaned data were then compiled into a structured spreadsheet with key variables including street address, date of removal, month, and year, and formatted for compatibility with ArcGIS Pro in preparation for geocoding and spatial analysis.

Despite its strengths, this dataset carries several limitations that should be acknowledged. Most importantly, the data capture carcass removal events rather than the precise moment of collision or mortality, meaning there may be a temporal lag between when an incident actually occurs

III. GIS Methodology

Once the dataset was finalized, it was imported into ArcGIS Pro for mapping and spatial analysis. Because the data consisted of street addresses rather than geographic coordinates, the first step was geocoding, a process that converts text addresses into specific points on a map. ArcGIS Pro geocoding tools were used to place each of the 222 successfully identified incidents as individual points on the map of Montclair. This process was not without challenges. Some addresses in the invoices were incomplete, abbreviated, or formatted inconsistently, which caused certain records to not match correctly during the initial geocoding run. These cases were reviewed manually and corrected where possible to improve address accuracy. After revisions were made, a final geocoding run was completed on March 25, 2026. Of the 223 reported deer incidents in the dataset, 222 were successfully identified and spatially located, representing a geocoding success rate of 99.55 percent. Only one record remained unresolved. These results indicate a highly reliable spatial dataset and provide strong confidence that the final map accurately reflected the location of reported deer incidents.

With all incidents geocoded and placed on the map, the next step was cartographic design and visualization. A customized basemap was selected in ArcGIS Pro to improve readability and provide geographic context. The National Geographic Style basemap was chosen because it offers a clean street network, labeled neighborhoods, and natural landscape features while remaining visually professional for publication purposes. In addition, the World Hillshade layer was added beneath the basemap to provide subtle topographic shading, allowing the terrain and ridge features surrounding Montclair to remain visible, a particularly useful design choice given Montclair's location along the eastern slope of the Watchung Mountain ridgeline. To focus attention on the study area, the township boundary layer was added to visually frame the map, with surrounding municipalities remaining visible for geographic reference. Deer carcass incident points were then symbolized above the basemap so that each location could be clearly identified against roads and neighborhood features. Layout adjustments (including legend formatting, scale refinement, and map extent calibration) were made before exporting each map as a clean, publication-ready product.

Two types of maps were produced for this study. The first type, point maps, show the exact location of each deer carcass removal as an individual marker on the street map of Montclair. These maps were created for each year individually, as well as for a direct comparison between 2020 and 2025, allowing for a clear picture of how the geographic distribution of incidents has changed over time. The second type, kernel density heatmaps, take a different approach. Rather than showing individual points, kernel density estimation (KDE) calculates how concentrated the incidents are across different parts of the township and produces a color-coded surface that makes hotspots immediately visible. KDE was chosen because it reveals spatial patterns that can be hard to see when looking at individual data points alone, particularly in areas where incidents cluster closely together.

The color scheme used in the heatmaps was chosen to make the density patterns as clear and intuitive as possible. Bright yellow indicates the highest concentration of incidents — the core hotspot areas. Moving outward from those centers, the color transitions through orange and red as density decreases, and eventually fades to blue at the outer edges where incidents are sparse. The same color scale was applied consistently across all heatmaps to allow for direct comparison between years.

To examine how the spatial pattern of incidents has changed over time, the year-by-year point maps were compared alongside one another, and the 2020 and 2025 heatmaps were analyzed side by side. This comparative approach makes it possible to see not just that incidents have increased in number, but also how their geographic footprint has shifted and grown across the township. Through both the point maps and the heatmaps, a small number of recurring hotspot locations emerged clearly. Areas that showed up consistently across multiple years and accounted for a disproportionate share of all recorded incidents. These locations were then cross-referenced with the township's road network and the surrounding geography to better understand why incidents concentrate where they do.

The combination of point mapping and kernel density analysis provided a detailed and layered picture of the deer incident problem in Montclair — one that would not have been possible through the invoice data alone. By translating municipal records into spatial visuals, this GIS-based approach turns raw administrative data into something that can directly inform planning and policy decisions at the local level.

IV. Results and Findings

Between 2020 and 2025, the Township of Montclair recorded a total of 222 confirmed deer carcass removal events, representing a 117% increase over the 5-year study period. White-tailed deer (*Odocoileus virginianus*) are the dominant deer species throughout New Jersey, present across all 21 counties and documented as a major component of the state's landscape in both rural and suburban settings, making it reasonable to attribute all recorded incidents to this species in the absence of contradicting evidence (NJDEP, 2025).

The year-by-year breakdown reveals a clear and mostly upward trajectory, with one notable interruption. Incidents rose sharply from 23 in 2020 to 38 in 2021 (+65%), before plateauing at 39 in 2022 (+3%). A significant drop occurred in 2023, when incidents fell to 27 (-31%), representing the only year of meaningful decline across the entire study period. However, this decline proved short-lived. Incidents rebounded sharply to 45 in 2024 (+67%), and reached a new five-year high of 50 in 2025 (+11%), averaging approximately 18.5% growth per year across the full study period. The overall trend is consistent with broader patterns documented in the literature. Research has shown that DVCs are increasing nationally as deer populations expand and suburban development encroaches further into forested landscapes (Morelle et al., 2020). Montclair's trajectory mirrors this pattern at the municipal scale, with the data suggesting that without intervention, incident counts will continue to climb.

Year	Incidents	YoY Change
2020	23	—
2021	38	+65%
2022	39	+3%
2023	27	-31%
2024	45	+67%
2025	50	+11%

Table 1 — Deer Carcass Removal Incidents by Year, Montclair NJ (2020–2025)

Beyond the year-over-year numbers, the monthly distribution of incidents reveals an equally clear and consistent pattern. October stands out as the single most dangerous month by a significant margin, accounting for 35 total incidents across all years, more than any other month. October recorded the highest incident count in 4 out of 5 years studied, with 2025 producing a striking peak of 12 incidents in October alone. This is not coincidental. It aligns directly with the white-tailed deer rut (the annual breeding season) during which bucks dramatically increase their range and movement patterns, significantly raising the probability of road crossings (Found & Boyce, 2011). Research on DVCs consistently identifies fall rutting behavior as one of the primary biological drivers of seasonal collision spikes (Hubbard et al., 2004). Taken together, fall (September through November) accounts for 78 incidents(37% of all recorded incidents across the 5-year period) making it by far the most dangerous season for Montclair motorists. Winter (December through February) follows with 46 incidents, summer (June through August) with 49, and spring (March through May) with the fewest at 37. The monthly heatmap provides a year-by-year visualization of this seasonal pattern, illustrating how October consistently emerges as the peak month across the study period.

Month	Total Incidents
January	13
February	10
March	17
April	9
May	11
June	17
July	18
August	14
September	16
October	**35**
November	27

December	23

Table 2 — Total Deer Carcass Incidents by Month, All Years Combined (2020–2025)

Spatially, the 222 incidents are anything but evenly distributed across the township. Rather than spreading uniformly across Montclair's road network, incidents cluster persistently along a small number of corridors, particularly those running along the township's western edge, adjacent to the forested reservations of the Watchung Mountain ridgeline. The single most incident-prone location identified in this study is Upper Mountain Avenue, which recorded 53 confirmed incidents across the five-year period, accounting for 24% of all reported incidents in the township. Upper Mountain Avenue recorded incidents in every single year of the study, with annual counts ranging from 7 to 14. No other street comes close to this level of persistent activity. Upper Mountain Avenue's dominance (more than four times the incident count of the second-ranked street) points to a structural issue rather than random variation. The road runs directly along the western edge of the township, parallel to the forested slopes of the First Watchung Mountain and adjacent to Mills County Park, making it a natural and recurring crossing point for deer moving between forested habitat and residential neighborhoods."

Rank	**Street**	**Total Incidents**
1	Upper Mountain Ave	53
2	Highland Ave	13
3	South Mountain Ave	12
4	Valley Rd	11
5	Midland Ave	7

Table 3 — Top 5 Hotspot Streets, Montclair NJ (2020–2025)

The year-by-year street maps illustrate how this spatial distribution has evolved over time. In 2020, incidents were relatively sparse and scattered across the township, with no single corridor showing dominant clustering. By 2025, the pattern had shifted dramatically, with a pronounced concentration of incidents along the Upper Montclair and central Montclair corridors, and new incidents appearing further south along South Mountain Avenue and Harrison Avenue. The side-by-side comparison of 2020 and 2025 makes this geographic expansion particularly visible.

The kernel density heatmaps reinforce and deepen this picture. The 2020 heatmap shows a relatively diffuse pattern, with incident density spread across a broad north-south band along the western portion of the township, reflecting the limited and scattered nature of that year's 23 incidents By contrast, the 2025 heatmap reveals a dramatically more concentrated pattern, with one huge distinct high-intensity cores emerging. one centered on Upper Montclair along Upper Mountain Avenue, and a second cluster in central Montclair , connected by a corridor of moderate incident density running the length of the township's western edge. The bright yellow cores in the maps represent the highest incident densities recorded across the entire study period, with intensity radiating outward through orange and red before dissipating into the blue periphery. This shift from a diffuse to a concentrated spatial pattern between 2020 and 2025 is significant. It suggests that deer incidents in Montclair are not simply increasing in number, but becoming more geographically focused, with certain corridors bearing a growing and disproportionate share of the township's total incident burden.

Finally, the rising incident count carries a direct and growing financial cost for the township. Deer carcass removal services in Montclair are contracted to Space Farms Zoo & Museum, which bills the township's Department of Health and Human Services on a per-removal basis through the municipal purchase order system. In 2021, the contracted removal rate was $44.00 per deer. By December 2025, a single consolidated removal invoice totaled $297, covering multiple removal events across locations including Valley Road, North Mountain Avenue, Grove Street, Gordenhurst Avenue, and Alexander Avenue. As incident counts rise, so does the cumulative financial burden on the township. If incident counts continue to grow at the observed average rate of approximately 18.5% per year, both the frequency and the associated removal costs can be expected to increase substantially without targeted intervention, a financial reality that reinforces the case for proactive road safety measures addressed in Section VI of this paper.

V. Explaining the 2023 Decline and 2024–2025 Rebound

One of the more analytically interesting features of Montclair's five-year incident record is its trend. Rather than a straight upward line, the data shows a sharp rise from 2020 to 2022, a notable drop in 2023, and then a sharp rebound in 2024 that carried into a new five-year high in 2025. Understanding what drove the 2023 decline, and why it did not last, requires looking beyond the numbers themselves. While the data alone cannot point to a single definitive cause, several plausible explanations emerge when the incident record is considered alongside what is known about deer behavior, traffic patterns, and the broader context of the post-pandemic period. None of these explanations can be confirmed or ruled out with the data available for this study, but taken together, they offer a reasonable framework for interpreting the trend.

The first and perhaps most structurally significant hypothesis is that the long-term rise in Montclair's deer incident count reflects the steady growth of the local deer population over time. White-tailed deer populations in New Jersey have expanded considerably since the early twentieth century, driven in large part by the extirpation of natural predators such as wolves and cougars from the northeastern landscape. The extirpation of these predators lowered natural mortality rates and, combined with agricultural and horticultural practices that created a constant supply of high-quality food resources, contributed to sustained deer population growth across the region (Rutgers NJAES, n.d.). In the absence of effective natural predation, deer populations in suburban areas like Montclair are largely regulated by hunting and municipal management programs. Essex County has operated an active deer culling program at South Mountain, Hilltop Reservations, along with Eagle Rock Reservation since 2010, but access limitations and the patchwork nature of suburban land ownership make population control difficult to sustain at the scale needed to meaningfully reduce incident rates (Montclair Local, 2025). The temporary dip in 2023 could reflect a particularly effective management year, while the rebound in 2024 and 2025 may represent the population reasserting itself as deer move in from surrounding areas to fill any gaps left by culling efforts.

The second hypothesis relates to traffic volume and the lingering effects of the COVID-19 pandemic. The year 2020, the first year of this study's dataset, coincided with widespread mobility restrictions that dramatically reduced the number of vehicles on the road across the United States. Research published in Scientific Reports found that while traffic volumes declined significantly during the pandemic, wildlife-vehicle collision rates did not fall proportionally, suggesting that decreased traffic volume does not always translate directly into fewer wildlife-vehicle collisions (Abraham & Mumma, 2021). However, the gradual return of normal traffic volumes from 2021 onward could be a reasonable contributing factor to the overall upward trend observed in Montclair's data. Animal-vehicle collisions increased by 7.2% on roadways across the United States between July 2020 and June 2021 compared to the previous year, a trend consistent with the traffic normalization that followed the initial pandemic period (Shen et al., 2023). The relatively modest incident count in 2020, the lowest of any year in the dataset, may partly reflect suppressed traffic rather than a genuinely lower underlying collision risk, which would mean the 2021 spike was at least partly a return to baseline rather than a true increase.

The third hypothesis is that habitat fragmentation in and around Montclair is pushing deer closer to roads over time, gradually increasing the structural collision risk regardless of short-term fluctuations in population or traffic. Research has consistently shown that as natural habitats are subdivided by roads and development, deer and other wildlife are forced to navigate increasingly fragmented landscapes, crossing roads more frequently in search of food, shelter, and mates. Studies have shown that habitat fragmentation from expanding road networks and development can disrupt natural movement patterns and push deer into areas near roadways, elevating the risk of collisions (Understanding the drivers of deer-vehicle collisions in South Carolina, 2025). In Montclair's case, the concentration of incidents along the western edge of the township, the space between Mills County Park and the forested Watchung ridgeline, is consistent with this dynamic. As that interface becomes more built up over time, the structural pressure on deer to cross roads in that corridor would be expected to increase.

The fourth and final hypothesis is more methodological than ecological: it is possible that some portion of the apparent growth in incidents reflects improvements in reporting and data collection over time rather than a true increase in the underlying rate of deer mortality. As the township's invoicing and record-keeping processes became more consistent and thorough across the study period, incidents that may have previously gone undocumented could have entered the record, inflating apparent growth. A related dimension of this same dynamic is public awareness. Residents who were previously unaware that the township offered deer carcass removal as a municipal service may have begun reporting incidents over time as word of the service spread. If a growing share of the community learned to call in these incidents rather than leaving them unreported, the dataset would reflect that expanding awareness as much as any real change in collision frequency. This is difficult to assess without access to internal administrative records or public outreach data beyond the invoices themselves, but it is worth acknowledging as a potential source of upward bias in the earlier years of the dataset.

Of these four hypotheses, the combination of long-term deer population growth and habitat fragmentation appears most consistent with the overall pattern of the data, both the sustained upward trend and the persistence of specific geographic hotspots across all five years. The COVID traffic normalization hypothesis offers a plausible explanation for the particularly low 2020 baseline, and improved reporting may account for some portion of early growth (Shen et al., 2023; Abraham & Mumma, 2021). The 2023 dip remains the hardest to explain with confidence, and may ultimately reflect a combination of factors, an effective culling season, an unusually mild fall rut, or simple year-to-year variation, rather than any single cause. What the data does make clear is that the rebound in 2024 and 2025 was sharp and sustained, suggesting that whatever temporarily suppressed incidents in 2023 did not represent a structural shift in the underlying trend.

VI. Policy and Sustainability Discussion

The data presented in this study makes a clear case that deer-related incidents in Montclair are not a random or unpredictable problem; they are concentrated in specific places, peak during specific months, and have been growing at a consistent rate for most of the past 5 years. That level of spatial and temporal structure is actually good news from a policy standpoint, because it means interventions do not need to be applied township-wide to be effective. Targeted, evidence-based measures focused on the right locations at the right times of year have the potential to meaningfully reduce both the safety risks and the financial burden that these incidents place on the township. Before discussing what is likely to work in Montclair's specific context, it is worth briefly addressing what the research says does not work, because the most commonly deployed deer safety measures are often among the least effective.

Standard static deer crossing signs, the familiar yellow diamond signs with a leaping deer, are the most widely used form of DVC mitigation in the United States, largely because they are inexpensive and easy to install. However, research has consistently found that standard passive signage has little to no measurable effect on either vehicle speeds or collision rates (Hedlund et al., 2004). Deer whistles, which are small devices mounted on vehicles intended to emit a sound that deters deer from the road, have similarly been found to be ineffective and are not supported by scientific evidence (Hedlund et al., 2004). Roadside reflectors, another commonly used low-cost measure, appear to have little long-term effect, with any initial impact eroding quickly over time (Hedlund et al., 2004). The takeaway from the literature is straightforward: there is no quick, cheap fix for DVCs, and the measures most commonly deployed at the municipal level are often the ones with the least evidence behind them. Montclair would be poorly served by investing in these approaches without complementing them with more substantive interventions.

What the research does support, strongly and consistently, is the combination of physical infrastructure and targeted, season-specific signage. On the infrastructure side, wildlife crossing structures paired with exclusion fencing represent the gold standard for DVC reduction. Studies have documented collision reductions of 80% or more where crossing structures and fencing have been properly designed and maintained (Huijser et al., 2016). A study on US Highway 64 in North Carolina found that three wildlife underpasses with associated fencing resulted in approximately 58% fewer white-tailed deer mortalities on that road segment compared to adjacent sections without such infrastructure (Rytwinski et al., 2016). While large-scale wildlife overpasses may not be immediately feasible on Montclair's residential streets, the principle is directly applicable: physical measures that guide deer to safer crossing points, away from high-traffic areas, are the most reliable long-term solution available.

Underlying all of these interventions is a basic ecological reality that any effective policy must reckon with: the vast majority of deer-vehicle collisions occur in darkness. Research consistently shows that DVCs occur predominantly at night, with one large-scale study finding collisions to be fourteen times more frequent in the two hours after sunset than in the two hours before it (Cunningham et al., 2022). This pattern points to driver visibility, not deer activity alone, as the dominant cause of collisions. Deer are equally active in the hours on either side of sunrise and sunset; what changes after dark is the driver's ability to see them. Hedlund et al. (2004) similarly note that DVCs occur predominantly in darkness, particularly where forest cover is close to the roadway. Seasonal changes in deer activity during the fall rut compound this risk further, as increased deer movement overlaps with longer periods of low-light driving. In Montclair's context, where the highest-incident corridor runs directly along a forested ridgeline, this overlap of low-light conditions and dense woodland edge habitat is not incidental. Any intervention strategy that fails to address the visibility dimension of the problem will be working against the underlying physics of the collision risk.

For Montclair specifically, Upper Mountain Avenue demands the most urgent attention. With 53 confirmed incidents over five years (24% of all township incidents) and recurring activity in every single year of the study period, this corridor has demonstrated a structural and persistent collision problem that is unlikely to resolve on its own. The road runs along the western edge of the township directly between the forested slopes of the First Watchung Mountain and Mills County Park, making it a natural and recurring deer crossing point. A targeted assessment of Upper Mountain Avenue's most incident-dense stretches, with a view toward installing deer exclusion fencing, reflective barriers, or guided crossing infrastructure at key points, would be a logical and evidence-backed first step. Highland Avenue, South Mountain Avenue, and Valley Road, the next three highest-incident streets, should be considered for secondary intervention once Upper Mountain Avenue is addressed.

On the signage front, the research does support one form of sign-based intervention that is meaningfully different from standard static signs: temporary, seasonal warning systems deployed specifically during the fall rut. A study by Sullivan et al. (2004) found that temporary warning signs equipped with reflective flags and flashing amber lights reduced DVCs by 50% during peak migration and rut periods, though the effect diminished in the second year, suggesting that sign placement should be rotated and refreshed to maintain driver attention. More promisingly, a study conducted in an urban environment directly comparable to Montclair found that warning signs installed specifically at DVC hotspot locations significantly reduced collisions at those sites, with a $4,000 signage investment generating an estimated $49,628 reduction in DVC-related costs (Boyce et al., 2011). The cost-effectiveness of this approach is particularly relevant for a municipality like Montclair, where budgetary constraints may limit the scope of infrastructure investment in the short term. Thankfully, Montclair's data also points to a clear seasonal intervention window. October, November, and December together account for a disproportionate share of all incidents across the study period, with October alone recording 35 incidents, more than any other month. Deploying seasonal deer advisory messaging on existing changeable message signs along Upper Mountain Avenue and adjacent corridors during September through November each year would align directly with the peak risk window. Research on Virginia's interstate system found that seasonal deer advisory messages on changeable message signs produced statistically significant speed reductions and were estimated to save between $700,000 and $1.4 million over the service life of the signs (Donaldson & Kweon, 2019). While Montclair's road network is smaller in scale, the underlying principle, using dynamic, time-specific messaging to raise driver awareness during the highest-risk period, translates directly to the township context.

Taken together, the evidence points toward a layered approach for Montclair: physical infrastructure improvements on Upper Mountain Avenue as a long-term priority, complemented by seasonal hotspot-targeted signage as an immediate and cost-effective near-term measure. Neither intervention alone is sufficient, but together they address both the structural wildlife movement problem and the driver awareness gap that the data reveals. Given that incidents have more than doubled over the past five years and show no signs of stabilizing, the case for action is not only supported by the research, it is supported by Montclair's own data.

VII. Limitations

No study is without its constraints. The most fundamental limitation is that this study is based entirely on carcass removal records rather than direct collision data. A deer carcass removal documents the end result of a deer-vehicle incident, but it does not capture the full picture. Deer that are struck by vehicles but manage to leave the road before dying, deer whose carcasses are removed by private parties or go unreported, and incidents that occur on private property may all fall outside the municipal invoice system entirely. This means the true number of deer-vehicle incidents in Montclair is almost certainly higher than the 223 recorded in this dataset. The data should therefore be understood as a confirmed minimum rather than a complete count.

Related to this, the dataset does not capture the precise moment of each incident. Because the invoices record when a carcass was removed rather than when the collision occurred, there is an inherent time lag between an incident and its entry into the record. In most cases, this lag is likely short, a matter of days, but it introduces a degree of imprecision into the monthly and seasonal analysis. An incident that occurred in late October, for example, may not appear in the record until November if the removal was delayed, which could slightly affect the monthly distribution findings.

The study also lacks several contextual variables that research has identified as important predictors of DVC risk. Traffic volume data for Montclair's roads was not available for this analysis, meaning it is not possible to determine whether the higher incident counts on streets like Upper Mountain Avenue reflect genuinely higher deer crossing activity or simply higher vehicle exposure. Similarly, weather data, time of day, and road speed information were not incorporated into the analysis, all of which are known to influence collision patterns (Morelle et al., 2020; Hubbard et al., 2004). Future studies that integrate these variables with the incident data developed here would produce a more complete picture of the risk landscape across the township.

A related assumption embedded in this dataset also warrants acknowledgment: the study treats every municipally removed deer carcass as the result of a vehicle strike, but this cannot be confirmed in every case. Deer found on or near roadways may have died from other causes, including disease, predation, or natural mortality, and subsequently been discovered and reported for removal. While vehicle collisions are by far the most common cause of deer carcasses requiring municipal removal on public roadways, the invoices themselves do not record the cause of death, and a small number of incidents in the dataset may not reflect true deer-vehicle collisions. This introduces a modest upward bias in the incident count that cannot be quantified without veterinary or field verification data.

Finally, while the GIS methodology used in this study provides a strong spatial framework for identifying hotspots and trends, geocoding address-level data always carries some degree of locational uncertainty. Addresses that were incomplete or inconsistently recorded in the original invoices required manual interpretation during the geocoding process, and it is possible that a small number of incidents were mapped to locations that do not precisely match where the removal actually took place. Every effort was made to verify and correct these cases, but some residual imprecision in the spatial data should be acknowledged.

Despite these limitations, the dataset represents one of the most consistent and verifiable sources of deer incident data available at the municipal scale in New Jersey. The three-layer verification structure of address, service record, and financial transaction gives the data a level of traceability and internal consistency that is rarely achievable with self-reported DVC sources, and makes the findings presented here a reliable foundation for evidence-based municipal planning.

VIII. Conclusion

This study set out to answer three questions about deer-related incidents in Montclair, New Jersey: where they are happening, when they are happening, and what the township can do about them. Alongside the written analysis, this project produced three tangible deliverables for the Township of Montclair's Department of Sustainability: a verified five-year incident dataset compiled from municipal invoices; a suite of GIS-generated maps showing the spatial distribution and evolution of incidents across the township; and a comprehensive statistical analysis report summarizing year-over-year trends, monthly patterns, seasonal breakdowns, and street-level hotspot rankings in an accessible, ready-to-use format. Five years of municipal invoice data, 223 confirmed incidents, and a systematic GIS-based spatial analysis later, the answers to all three questions are clearer than they have ever been, and the tools to act on them are now in hand.

The “where” is Western Montclair, Especially Upper Mountain Avenue. With 53 incidents across every single year of the study period, this corridor accounts for nearly a quarter of all deer-related incidents in the township and stands apart from every other street by a significant margin. The spatial analysis makes clear that this is not a coincidence. Upper Mountain Avenue runs directly along the interface between Montclair's residential neighborhoods and the forested slopes of the First Watchung Mountain and Mills County Park, placing it squarely in the path of deer moving between natural habitat and developed land. The heatmaps produced through kernel density analysis reinforce this picture, showing a growing concentration of incident density along the township's western edge that has become more pronounced with each passing year.

The “when” is fall, October in particular. October alone accounts for 35 incidents across the six-year study period, more than any other month, and recorded the highest incident count in five out of six years. The biological driver behind this pattern is well established in the literature: the white-tailed deer rut dramatically increases deer movement and road-crossing behavior each fall, raising collision risk in a predictable and recurring window that runs from September through November (Found & Boyce, 2011; Hubbard et al., 2004). The consistency of this seasonal pattern across all six years of data means that Montclair's highest-risk period is not just identifiable in hindsight, it is predictable in advance, which has direct implications for when interventions should be deployed.

The “what” is a layered approach that matches the scale and structure of the problem. The research is clear that cheap, passive measures like standard deer crossing signs and deer whistles do not produce meaningful reductions in DVCs (Hedlund et al., 2004). What does work is a combination of physical infrastructure, fencing, crossing structures, and guided passage points, along with targeted seasonal signage deployed at known hotspot locations during the fall rut window. Upper Mountain Avenue is the logical starting point for both types of intervention, given its documented dominance in the incident record. Seasonal changeable message signs and hotspot-specific warning systems represent cost-effective near-term options, while a longer-term assessment of fencing and crossing infrastructure along the western corridor would address the structural wildlife movement issue that underlies the persistent hotspot pattern.

In addition to the written analysis, this study produced a suite of GIS-generated maps, including year-by-year incident point maps, a direct 2020-to-2025 comparison, and kernel density heatmaps for key years of the study period, that are provided to the Township of Montclair's Department of Sustainability as standalone planning documents. These maps translate five years of administrative records into an immediately usable visual resource: township staff can identify high-priority corridors at a glance, track how the spatial pattern of incidents has shifted over time, and use the hotspot data to inform decisions about where to focus outreach, signage, or infrastructure assessment. The maps are designed to remain useful as a baseline against which future years of data can be compared, allowing the township to monitor whether interventions are producing measurable changes in the geographic distribution of incidents.

Beyond the specific findings, this study demonstrates something important about the value of municipal administrative data. Deer carcass removal invoices are not typically thought of as research material, they are operational records generated in the course of routine municipal services. But when compiled, geocoded, and analyzed systematically, they reveal spatial and temporal patterns that are directly actionable for local planning and sustainability efforts. The three-layer verification structure of address, service record, and financial transaction makes this dataset more reliable than most DVC data sources available in the literature, and the five-year longitudinal span makes it possible to identify trends that a single-year snapshot would miss entirely.

Montclair's deer incident problem has more than doubled in five years. It is geographically concentrated, seasonally predictable, and growing. The data and the research both point in the same direction; it points toward targeted, evidence-based intervention before the trend continues to accelerate. This study provides the spatial and temporal foundation that the Township of Montclair's Department of Sustainability needs to move from awareness to action.

References

Cunningham, C. X., Nuñez, T. A., Hentati, Y., Sullender, B., Breen, C., Ganz, T. R., Kreling, S. E. S., Shively, K. A., Reese, E., Miles, J., & Prugh, L. R. (2022). Permanent daylight saving time would reduce deer-vehicle collisions. Current Biology, 32(22), 4982–4988. https://doi.org/10.1016/j.cub.2022.10.007

Essex County Parks. (2025a). Eagle Rock Reservation. https://essexcountyparks.org/parks/eagle-rock-reservation

Essex County Parks. (2025b). South Mountain Reservation. https://essexcountyparks.org/parks/south-mountain-reservation/about

Found, R., & Boyce, M. S. (2011). Predicting deer-vehicle collisions in an urban area. Journal of Environmental Management, 92(10), 2486–2493. https://doi.org/10.1016/j.jenvman.2011.05.010

Hedlund, J. H., Curtis, P. D., Curtis, G., & Williams, A. F. (2004). Methods to reduce traffic crashes involving deer: What works and what does not. *Traffic Injury Prevention*, *5*(2). https://www.defenders.org/sites/default/files/publications/methods_to_reduce_traffic_crashes_involving_deer.pdf

Hubbard, M. W., Danielson, B. J., & Schmitz, R. A. (2004). Deer-vehicle collisions: An understanding of accident characteristics and driver knowledge, attitudes, and behaviors. National Transportation Library. https://rosap.ntl.bts.gov/view/dot/26774

Insurance Information Institute. (2025). Facts + statistics: Deer vehicle collisions. https://www.iii.org/fact-statistic/facts-statistics-deer-vehicle-collisions

Morelle, K., Marchais, T., Nguyen, T., & Lejeune, P. (2020). Wildlife-vehicle collisions: Influencing factors, data collection and research methods. Biological Conservation, 251, 108758. https://doi.org/10.1016/j.biocon.2020.108758

NJDEP Division of Science and Research. (2025). Wildlife populations: White-tailed deer. New Jersey Department of Environmental Protection. https://dep.nj.gov/dsr/environmental-trends/white-tailed-deer/

State Farm. (2025). New State Farm® data reveals fewer animal collisions, but autumn months remain most dangerous. https://newsroom.statefarm.com/new-state-farm-data-reveals-fewer-animal-collisions-but-autumn-months-remain-most-dangerous/

Abraham, J. O., & Mumma, M. A. (2021). Elevated wildlife-vehicle collision rates during the COVID-19 pandemic. *Scientific Reports, 11,* 20391. https://www.nature.com/articles/s41598-021-99233-9

Boyce, M. S., Bleiker, K., & Haynes, G. (2011). Warning signs mitigate deer-vehicle collisions in an urban area. *Human-Wildlife Interactions.* https://www.researchgate.net/publication/230082047

Donaldson, B. M., & Kweon, Y. J. (2019). Effectiveness of seasonal deer advisories on changeable message signs as a deer crash mitigation tool. *Transportation Research Record.* https://journals.sagepub.com/doi/10.1177/0361198118823204

Huijser, M. P., Clevenger, A. P., & McGowen, P. (2016). Effectiveness of short sections of wildlife fencing and crossing structures along highways in reducing wildlife-vehicle collisions and providing safe crossing opportunities for large mammals. *Biological Conservation.* https://www-sciencedirect-com.ezproxy.montclair.edu/science/article/pii/S0006320716300350?via%3Dihub

Montclair Local. (2025, January 6). Essex County details deer hunt plans for 2025. https://montclairlocal.news/2025/01/essex-county-details-deer-hunt-plans-for-2025/

Rutgers NJAES. (n.d.). An overview of white-tailed deer status and management in New Jersey. *Rutgers New Jersey Agricultural Experiment Station.* https://njaes.rutgers.edu/fs1202/

Rytwinski, T., Soanes, K., Jaeger, J. A. G., Fahrig, L., Findlay, C. S., Houlahan, J., van der Ree, R., & van der Grift, E. A. (2016). Effectiveness of wildlife underpasses and fencing to reduce wildlife-vehicle collisions. *USGS.* https://wildlife-onlinelibrary-wiley-com.ezproxy.montclair.edu/doi/epdf/10.2193/2009-535

Shen, G., Lin, J., & Liu, Z. (2023). A descriptive analysis of deer-vehicle collisions along Missouri interstate highways. *Transportation Research Interdisciplinary Perspectives.* https://www.sciencedirect.com/science/article/pii/S2666719323000419

Sullivan, T. L., Williams, A. F., Messmer, T. A., Hellinga, L. A., & Kyrychenko, S. Y. (2004). Effectiveness of temporary warning signs in reducing deer-vehicle collisions during mule deer migrations. *Wildlife Society Bulletin, 32(3).* https://www.researchgate.net/publication/232673823

Understanding the drivers of deer-vehicle collisions in South Carolina. (2025). *Global Ecology and Conservation.* https://www.sciencedirect.com/science/article/pii/S2351989425004585